

Hawkeye for Amateurs

Section 1: Statement of the Problem (The Whys)

Subjective decisions in sport are heartbreaking and can turn the course of many games. As a player myself, I have been harmed by these wrong decisions on countless occasions, as have my opponents. I know many people, including myself, who would love the idea of a review system in their amateur leagues. This app can improve decisions in cricket and, if successful, can be expanded to other sports in the future.

This problem is apparent for all non-professional cricket players, especially in a decision called Leg Before Wicket (LBW), which uses the umpire's judgment to call on whether or not the ball would have hit a target, if the leg did not block the path of the ball, based on some criteria. Although technology to predict this path exists, it is exclusively used for professional cricket, preventing us (amateurs) from utilizing it. Additionally, professional review systems require multiple, precise setups to improve accuracy. Most of us don't have access to such fancy equipment; however, my goal is to maximize accuracy in my app, despite the limitations.

My goal is to objectify the LBW decision using technology and AI, such as ball-tracking, for the general public to use in their amateur-level games. I plan to create a published app or program that can accurately predict a ball's path for an LBW decision with one camera.

My goal is to accurately objectify the LBW decision with one or two cameras, for amateur cricketers, by using computer vision and AI to create a successful ball-path prediction (Hawkeye).

Section 2: Process (Materials, Preparations, and Steps)

My project doesn't require many external materials. In fact, part of the problem statement is to create this prediction with minimal external hardware. I plan to use a regular phone camera, in a calibrated position, and customized software to do this. With an intermediate coding background, yet a surface-level understanding of AI's mechanics, I plan to do a lot of research as well as consult with experts in this field.

My father, who works frequently on computer vision and ML tasks, is a perfect candidate stakeholder for this project. I also plan to look into experts in the field of Hawkeye who have actually made the professional app.

If my program can be created by the deadline, then I will create an app to publish to the general public so that players and I can use the app to incorporate a review system into our games.

Plan:

1. First, I need to create plans for my research. I have to be very thorough in the world of AI and computer vision to create a project of this calibre. I plan to use technology that has been developed, as well as new software.
 - a. This phase would be a big part of the project as it will discuss feasibility and limitations
2. Then, I plan to start drafting a program that incorporates these elements. This step would probably integrate with step 1, as I will research while coding their program.
 - a. Need feedback for it
3. If a program is created in time, then I will work on developing an app with a customized UI for this program.
 - a. This part will take a long time, as I need to research app development basics as well as find a way to seamlessly integrate my program into the app with a usable UI.
4. If an app is created, then I plan to publish the app for public use!

What next?

I can then explore avenues for reviewing systems of other sports. Since most review systems are exclusive to professional games, my app will create a way for other sports that require review systems (that are automated and don't require umpires/referees), like tennis.

Section 3: Goals (Final Product)

The final product will likely be in the form of a program or an app. If the project can be completed due to feasibility issues, it will likely be a research project on Hawkeye and computer vision. The product will accurately predict a ball's path after obstruction to objectify an LBW decision. If created into a published app, the app will contain the program as well as UI that eases the use of the program.

My app/program will receive feedback and hopefully be used by many to create a review system. I hope to spread my app such that it is used across the Bay Area to prevent bad umpiring calls, specifically the LBW decision, from ruining cricket games.

Measured by key criteria, several success factors will characterize this project. The first is to develop an advanced app or program that can accurately predict LBW decisions in cricket, making the traditionally subjective call more objective and data-driven. This system will employ state-of-the-art technology and exact algorithms to consider trajectory, impact, and other essential aspects of the ball. Secondly, besides its technical correctness, the capability for expansion and distribution of such an app or program to ensure it will be easily accessible for cricket players and enthusiasts all over the world will also be a key success factor. In essence, what this should lead to is a reliable tool that introduces consistency and fairness into the game.